Antoine Franchini

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Antoine Franchini antoineinria

Experience

PhD: Confidence interval construction for extreme quantiles — *Grenoble, France*

Inria & EDF — *Dec 2024 – Dec 2027*

- Refinement of the asymptotic distributions of quantile estimators from the Extreme Value Theory.
- Implementation of machine learning methods (LDA, SVM, HDDA, Neural network) to classify the data in order to select the 'best' confidence interval.
- Development of alternative quantile estimators using Peak Over Threshold method.

Research Internship — *Grenoble, France*

Inria & EDF — *May 2024 – Sep 2024*

- Review of the different quantile estimation methods from the Extreme Value Theory.
- Study of the asymptotic distributions of the Weissman estimator in the heavy-tail context.
- Preparation of the PhD project.

Education

Université Paris Cité — *Sep 2022 – Oct 2024*

Master degree: Random modeling in finance

- **Coursework:** R and C++ languages, stochastic calculus, financial derivatives, statistical learning, point processes, time series.

Université Paris Cité — *Sep 2019 – Jun 2022*

Bachelor degree in general and applied mathematics

Projects

(FR) : Prédiction des Droits de Mutation à Titre Onéreux (DMTO) - *Hexadone* — *Grenoble, France, May 2025*

SEMIES 2025, (FR) Website

- Social and financial data investigations to predict the DMTOs.
- Establishment of a catalogue of prediction methods: Random forest, Prophet by Meta, Autoregressive models.

Teaching

Practical courses in statistics in R for 2nd year of Biolgy Bachelor — *Saint-Martin d'Hères, France*

Université Grenoble Alpes

Sep 2026 – Dec 2026
Sep 2025 - Dec 2025

Algebra and analysis courses for 1st year of Physics Bachelor — *Saint-Martin d'Hères, France*

Université Grenoble Alpes

Jan 2025 – May 2025

Publications

Adaptive confidence intervals for extreme quantiles from heavy-tailed distributions — hal.science/hal-05322341
Antoine Franchini, Stéphane Girard, Anne Dutfoy — *Jul 2026*
Statistics & Computing

Talks

(FR) Sélection d'intervalles de confiance pour les quantiles extrêmes de lois à queue lourde — *hal.science/hal-05650334*

- *JDS 2026, Clermont-Ferrand, France.* Website — *Jun 2026*

Adaptive procedure for confidence intervals for extreme quantiles from heavy-tailed distributions — *hal.science/hal-05538621*

- *CJC-MA 2026, Champs-Sur-Marne, France.* Website — *Mar 2026*

Quantification of extrapolation performances of extreme quantile estimators — *hal.science/hal-04850514*

- *Valpred 2024, Aussois, France.* Website — *Dec 2024*

Skills

Computer Languages: R, Python.